



Chapter 6

JavaScript: Introduction to Scripting

Internet & World Wide Web
How to Program, 5/e

<ftp://111webdesign:webdesign@120.114.190.22:4000>





OBJECTIVES

In this chapter you will:

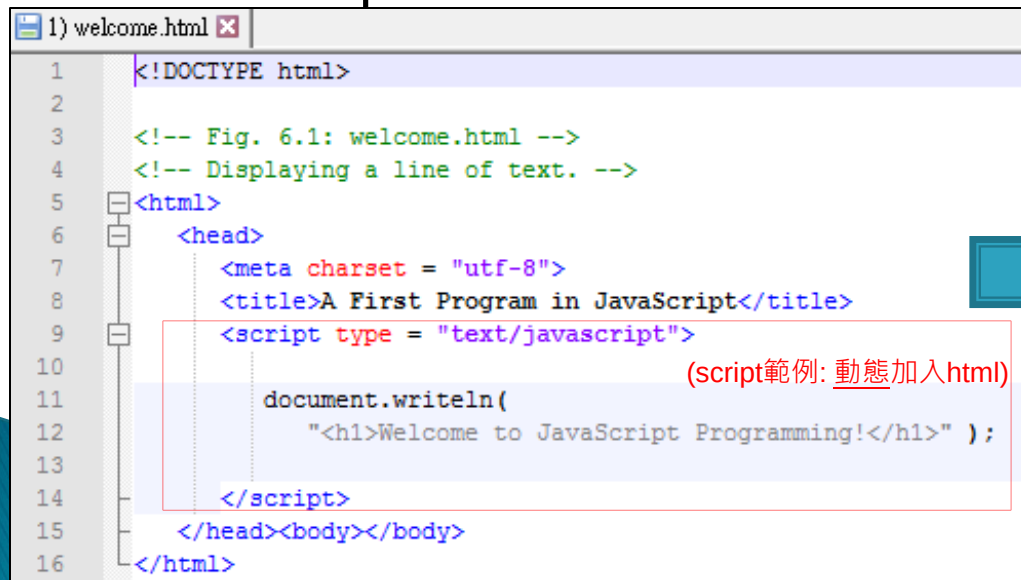
- Write simple JavaScript programs.
- Use input and output statements.
- Learn basic memory concepts.
- Use arithmetic operators.
- Learn the precedence of arithmetic operators.
- Write decision-making statements to choose among alternative courses of action.
- Use relational and equality operators to compare data items.

6.1 Introduction

- ▶ **JavaScript** (可執行一系列動作)
 - Scripting language (手稿/腳本語言) which is used to enhance the **functionality** (功能性) and **appearance** (外觀) of web pages.
 - 註：JavaScript主要用來增強網頁功能性及外觀 (**動態**)。
(否則html一般為靜態)
- ▶ Before you can run **code examples** with JavaScript on your computer, you may need to **change** your browser's **security settings**.
 - Firefox, Chrome, Opera, Safari and the Android browser 預設啟用.
 - IE 預設不啟用 **script**.

6.2 Your First Script: Displaying a Line of Text with JavaScript in a Web Page

- ▶ First, we show a **simple script** that displays the text "**Welcome to JavaScript Programming!**" in the HTML5 document.
 - All **major (主流) web browsers** contain **JavaScript interpreters (直譯器)**, which process the **commands** written in JavaScript.
 - Example:



```
1 <!DOCTYPE html>
2
3 <!-- Fig. 6.1: welcome.html -->
4 <!-- Displaying a line of text. -->
5 <html>
6   <head>
7     <meta charset = "utf-8">
8     <title>A First Program in JavaScript</title>
9     <script type = "text/javascript">
10
11       document.writeln(
12         "<h1>Welcome to JavaScript Programming!</h1>" );
13
14     </script>
15   </head><body></body>
16 </html>
```

(script範例: 動態加入html)

(動態執行)

(動態印出)





6.2 Your First Script: Displaying a Line of Text with JavaScript in a Web Page (cont.)

- ▶ Usually, **JavaScripts** appear in the **<head>** section of the HTML5 document.
 - The browser **interprets** the contents of the **<head>** section first
 - 註：放在<head>用於預先裝載，放在<body>則用於呈現頁面內容

<script>...</script> 描述腳本內容

- ▶ The **<script>** tag indicates to the browser that the text that follows is **part of a script**.
 - **type = “...”** 型別屬性
 - **Attribute type** (屬性型別) specifies the scripting language used in the script—such as **type= “text/javascript”**
 - 註：也可以是其他類script，如：VBscript **type=“text/vbscript”**

Ps. 若未加 **type = “text/javascript”**，瀏覽器仍可以run，因為現今瀏覽器很聰明。



6.2 Your First Script: Displaying a Line of Text with JavaScript in a Web Page (cont.)

- ▶ specifies the **MIME script type** (腳本型態) and the **scripting language** (腳本語言)
- ▶ 註：**MIME (Multipurpose Internet Mail Extensions, 多用途網際網路郵件擴展)** 為一種網際網路標準，其擴展電子郵件標準，能支援 非文本格式附件 (二進位、聲音、圖像等)。
- ▶ 註：HTML5的預設script type為 **type="text/html"**.
- ▶ **A string of characters (字元串)** can be contained between **double quotation mark ("...")**, **雙引號** or **single quotation mark ('...')**, **單引號**
//兩者都行
- ▶ **document 文檔物件**
 - 代表HTML5文檔在瀏覽器的呈現，允許script存取HTML5文檔內容
 - 註：物件(object) 含有attributes (存放data) 及 behaviors (methods函式)，其儲存在電腦端的記憶體內



6.2 Your First Script: Displaying a Line of Text with JavaScript in a Web Page (cont.)

▶ 物件用法：

物件名稱.方法名稱("參數1", "參數2", ...);

Example:

document.writeln("...");

```
<script type = "text/javascript">
    document.writeln(
        "<h1>Welcome to JavaScript Programming!</h1>" );
</script>
```

(script範例: 動態加入html)

- ▶ The **parentheses** (括弧) following the **name of a method** (方法名稱) contain **the arguments** (參數) that the method requires to perform its task
- ▶ Every **statement** should end with a **semicolon** (";", 分號)
- ▶ JavaScript is **case sensitive** (嚴格大小寫)

(類似c語言的printf)

.writeln("..."); //功能：寫入一行html

- Writes **a line** of HTML5 text in the HTML5 document (寫入一行文字至HTML5)
- Browser will interpret the HTML5 elements after the line of HTML5 text is written in the document



6.2 Your First Script: Displaying a Line of Text with JavaScript in a Web Page (cont.)

- ▶ **JavaScript code** can be placed in a **separate file** (放在獨立的檔案)
- ▶ To be **included** in the HTML5 document (再透過HTML5文檔引用).
- ▶ This makes the code **more reusable** (重覆利用)
 - ▶ To be the **JavaScript libraries** (函式庫).
 - ▶ 註：形成**函式庫集**，讓其他HTML5檔也能使用。
 - ▶ Example:

(.js檔)

welcome.html

```
-----
<!DOCTYPE html>

<!-- Fig. 6.1: welcome.html -->
<!-- Displaying a line of text. -->
<html>
  <head>
    <meta charset = "utf-8">
    <title>A First Program in JavaScript</title>
    <script src="helloWorld.js">
    </script>
  </head> <body> </body>
</html>
```

(指定src來源)

helloworld.js

```
-----
document.writeln("<h1>Welcome to JavaScript
Programming!</h1>");
```


6.3 Modifying Your First Script

- ▶ Another example:
 - using the **CSS color** property (加上有色字體).
 - “+” operator (稱為“concatenation operator”，連接兩字串)

```
3) welcome2.html x
1  <!DOCTYPE html>
2
3  <!-- Fig. 6.2: welcome2.html -->
4  <!-- Printing one line with multiple statements. -->
5  <html>
6  <head>
7      <meta charset = "utf-8">
8      <title>Printing a Line with Multiple Statements</title>
9      <script>
10         (write範例: 繼續寫入)
11         document.write( "<h1 style = 'color: magenta'>" );
12         document.write( "Welcome to JavaScript " +
13             "Programming!</h1>" );
14         (連接二字串)
15     </script>
16 </head><body></body>
17 </html>
```

(可分成多個write來寫)



6.3 Modifying Your First Script (Cont.)

(物件.方法)

window.alert("..."); //跳出文字對話框

Displaying Text in an Alert Dialog

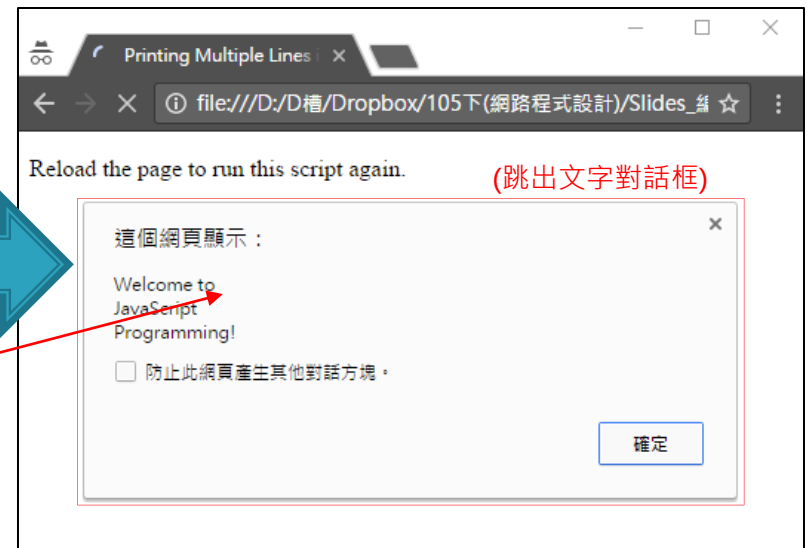
▶ Dialogs

- To “pop up” (彈出) information on the screen to grab the user’s attention (吸引讀者目光).
- 註：使用瀏覽器window物件的alert方法來呈現。
- 註：將要出現的“文字”資訊，作為alert方法的參數(“...”)。

* (課外補充) 若想認識更多函式，可參考線上資訊：

https://www.w3schools.com/js/js_window.asp

```
1 <!DOCTYPE html>
2
3 <!-- Fig. 6.3: welcome3.html -->
4 <!-- Alert dialog displaying multiple lines. -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>Printing Multiple Lines in a Dialog Box</title>
9   <script>
10
11     window.alert( "Welcome to\nJavaScript\nProgramming!" );
12
13   </script>
14 </head>
15 <body>
16   <p>Reload the page to run this script again.</p>
17 </body>
18 </html>
```



6.3 Modifying Your First Script (Cont.)

Escape Sequences //跳脫序列 (特殊功能字元)

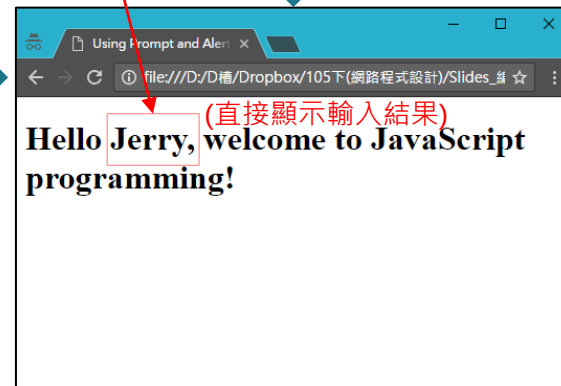
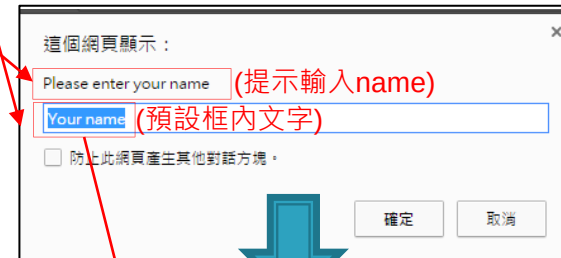
- ▶ Combining a **backslash** (“\”, 反斜線) with the **next character** (下一個字元) to form an **escape sequence**.
- ▶ Ex: “\n” is the **newline character** (換行字元).

Escape sequence	Description
\n	<i>New line</i> —position the screen cursor at the beginning of the next line. (換行)
\t	<i>Horizontal tab</i> —move the screen cursor to the next tab stop. (往前一個tab空格)
\\	<i>Backslash</i> —used to represent a backslash character in a string. (代表出現單個“反斜線”字元)
\"	<i>Double quote</i> —used to represent a double-quote character in a string contained in double quotes. For example, <code>window.alert("\"in double quotes\"");</code> (代表一個雙引號) displays "in double quotes" in an alert dialog.
\'	<i>Single quote</i> —used to represent a single-quote character in a string. For example, (代表一個單引號) <code>window.alert("'"in single quotes'");</code> displays 'in single quotes' in an alert dialog.

6.4 Obtaining User Input with prompt Dialogs

- ▶ **Scripting 功用** (動態產生, ex: 打招呼)
 - To **dynamically** generate part of web content (動態產生web內容)
 - Example:
 - To show a **dynamic welcome page** that obtains the **user's name**, then **displays** it on the page.
- 變數 = **window.prompt(“參數1”, “參數2”)**; 文字提示對話框
 - To use a **prompt** dialog to let users **enter some information** (讓使用者輸入“資訊”).
 - 參數1: 提示使用者文字框上的資訊
 - 參數2: 文字框內提示文字
 - “=”: To **assign a value** from right part.

```
5) welcome4.html x
1 <!DOCTYPE html>
2
3 <!-- Fig. 6.5: welcome4.html -->
4 <!-- Prompt box used on a welcome screen -->
5 <html>
6 <head>
7 <meta charset = "utf-8">
8 <title>Using Prompt and Alert Boxes</title>
9 <script>
10
11     var name; // string entered by the user
12
13     // read the name from the prompt box as a string
14     name = window.prompt( "Please enter your name", "Your name" );
15
16     document.writeln( "<h1>Hello " + name +
17                       ", welcome to JavaScript programming!</h1>" );
18
19 </script>
20 </head><body></body>
21 </html>
```



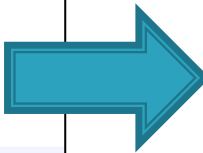
6.4.1 Dynamic Welcome Page (cont.)

- ▶ **Keywords (專用字詞)** are words with **special meaning (特殊涵意)** in JavaScript
- ▶ **var ... , ... , ... ; //宣告變數名稱**
 - To declare the **names of variables** (跟c語言類似)
 - 註：Client端電腦會預留每個變數記憶體，供儲存和讀取。
 - 註：*每個變數使用前都必先宣告
 - 註：變數命名規則
 - 允許包含 **letters (字母)**, **digits (數字)**, **underscores (“_” 底線)** 及 **dollar signs (“\$” 錢字符)**
 - 不允許數字開頭、不可使用JavaScript 專用字詞.
- ▶ 註：分號結束(“;”)、多變數宣告可用逗號(“,”)分開
- ▶ **Comments (註解)**
 - **//...** : 單行註解
 - **/* ... */** : 多行註解 (跟c語言類似)
- ▶ **null //專用字詞：變數初始化使用**
 - Ex: number = **null**; //Means a variable **without value**

6.4.1 Dynamic Welcome Page (cont.)

(註：var未指定型態，會根據寫入內容決定型態)

- ▶ **parseInt(字串變數);** //函式：字串轉數字
 - converts a string to an integer
- ▶ 註：“+”：
 - for two string concatenation (兩字串連接)
 - for two value addition (兩數字相加)



```
6) addition.html x
1 <!DOCTYPE html>
2
3 <!-- Fig. 6.7: addition.html -->
4 <!-- Addition script. -->
5 <html>
6   <head>
7     <meta charset = "utf-8">
8     <title>An Addition Program</title>
9     <script>
10
11     var firstNumber; // first string entered by user
12     var secondNumber; // second string entered by user
13     var number1; // first number to add
14     var number2; // second number to add
15     var sum; // sum of number1 and number2
16
17     // read in first number from user as a string
18     firstNumber = window.prompt( "Enter first integer" );
19     //讀入字串
20     // read in second number from user as a string
21     secondNumber = window.prompt( "Enter second integer" );
22     //讀入字串
23     // convert numbers from strings to integers
24     number1 = parseInt( firstNumber ); (將字串轉數字)
25     number2 = parseInt( secondNumber );
26
27     sum = number1 + number2; // add the numbers
28
29     // display the results (數字相加)
30     document.writeln( "<h1>The sum is " + sum + "</h1>" );
31
32   </script>
33 </head><body></body>
34 </html>
```



這個網頁顯示：

Enter first integer (提示輸入數字1)

7

☐ 防止此網頁產生其他對話方塊。

確定 取消



這個網頁顯示：

Enter second integer (提示輸入數字2)

4

☐ 防止此網頁產生其他對話方塊。

確定 取消

An Addition Program x

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The sum is 11 (輸出計算結果)

6.5 Memory Concepts

- ▶ Every **variable** (變數) has a **name** (名稱), a **type** (型別) and a **value** (值), stored in **the memory** (在記憶體上預留空間).
- ▶ **變數值寫入** : a **value** is placed in a **memory** (記憶體) **location**, the value replaces the previous value in that location.
- ▶ **變數值讀出** : a **value** is **read out** of a memory location

	(記憶體內容)
number1	45

Fig. 6.8 | Memory location showing the name and value of variable number1.

number1	45
number2	72

Fig. 6.9 | Memory locations after inputting values for variables number1 and number2.

number1	45
number2	72
sum	117

Fig. 6.10 | Memory locations after calculating the sum of number1 and number2.



6.5 Memory Concepts (Cont.)

- ▶ **變數型態** : no need to declare **variable's type** before they are used in a script (**免宣告型態**) (會根據寫入內容，來決定型態)
- ▶ **A javascript variable** can contain **a value of any data type** (數值可是任何型態)
 - ▶ **loosely typed** language (較寬鬆型態之語言)
 - ▶ a variable **is declared** without **giving a value**, it can not be use (**logic error**).
 - 註：可先給變數 **null** 值，來表示此變數未設值。

6.6 Arithmetic

- ▶ 基本運算符號：“+”，“-”，“*”，“/”，“%” are **binary operators**, because they each operate on **two operands**
 - 註：二元運算子，提供兩運算元作計算
 - 註：% 表示除法取餘數
 - 註：計算通常不換行

JavaScript operation	Arithmetic operator	Algebraic expression	JavaScript expression
Addition	+	$f + 7$	<code>f + 7</code> (加法)
Subtraction	-	$p - c$	<code>p - c</code> (減法)
Multiplication	*	bm	<code>b * m</code> (乘法)
Division	/	x/y or $\frac{x}{y}$ or $x \div y$	<code>x / y</code> (除法)
Remainder	%	$r \bmod s$	<code>r % s</code> (除餘)

Fig. 6.1 Arithmetic operators.

6.6 Arithmetic (Cont.)

- ▶ **Parentheses (括弧)** can be used to group expressions as in algebra.
- ▶ **Operators precedence (運算有優先順序)** :
 - **Multiplication (乘法)**, **division (除法)** and **remainder (除餘)** operations are **applied first (優先)**.
 - Operators are applied from **left to right (由左至右計算)** if containing **several** operations.
 - 規則：先“乘除”後“加減”
 - **Multiplication (乘法)**, **division (除法)** and **remainder (除餘)** operations are said to have the **same level of precedence (乘除相同優先序)**.
 - **Addition (加法)** and **subtraction (減法)** operations are applied next, and have the **same level of precedence (加減相同優先序)**.

	Operator(s)	Operation(s)	Order of evaluation (precedence)	
(高優先權)	*, / or %	Multiplication Division Remainder	Evaluated first. If there are several such operations, they're evaluated from left to right.	(由左至右)
(低優先權)	+ or -	Addition Subtraction	Evaluated last. If there are several such operations, they're evaluated from left to right.	

Fig. 6.6-1 Precedence of arithmetic operators.

6.7 Decision Making (決策): Equality and Relational Operators

- ▶ 條件判斷 `if(...)`
 - ▶ To **make a decision** based on the **True (成立)** or **False (不成立)** of a condition
 - **True (成立)** : 執行if內的statement
 - **False (不成立)** : 不執行if內的statement
 - ▶ **Conditions (判斷條件)** can be the **equality operators (等式, 例如 : 判斷是否相等)** and **relational operators (判斷關係, 例如 : 是否大/小於等於)**
 - The **equality operators** associate from **left to right (由左至右)**.

Standard algebraic equality operator or relational operator	JavaScript equality or relational operator	Sample JavaScript condition	Meaning of JavaScript condition
<i>Equality operators (等式 : 是否相等)</i>			
=	==	x == y	x is equal to y
≠	!=	x != y	x is not equal to y
<i>Relational operators (關係 : 比較大小)</i>			
>	>	x > y	x is greater than y
<	<	x < y	x is less than y
≥	>=	x >= y	x is greater than or equal to y
≤	<=	x <= y	x is less than or equal to y

Fig. 6.11 Equality and relational operators.

註：等式的優先序低於關係式 (當混合使用時).

6.7 Decision Making: Equality and Relational Operators (Cont.)

- ▶ Example: To display a **time-sensitive greeting** on a welcome page

- 根據client端時間：早/中/晚進行不同問候。

`var now = new Date();` // 日期物件

- 註：利用new產生新物件給變數"now"

```
1 <!DOCTYPE html>
2
3 <!-- Fig. 6.14: welcome5.html -->
4 <!-- Using equality and relational operators. -->
5 <html>
6 <head>
7   <meta charset = "utf-8">
8   <title>Using Relational Operators</title>
9   <script>
10
11     var name; // string entered by the user
12     var now = new Date(); // current date and time
13     var hour = now.getHours(); // current hour (0-23)
14
15     // read the name from the prompt box as a string
16     name = window.prompt( "Please enter your name" );
17
18     // determine whether it's morning (條件判斷)
19     if ( hour < 12 ) //中午之前 (早上)
20       document.write( "<h1>Good Morning, " );
21
22     // determine whether the time is PM
23     if ( hour >= 12 ) //中午之後 (下午)
24     {
25       // convert to a 12-hour clock
26       hour = hour - 12;
27
28       // determine whether it is before 6 PM
29       if ( hour < 6 )
30         document.write( " <h1>Good Afternoon, " );
31
32       // determine whether it is after 6 PM
33       if ( hour >= 6 ) //晚上
34         document.write( " <h1>Good Evening, " );
35     } // end if
36
37     document.writeln( name +
38       ", welcome to JavaScript programming!</h1>" );
39
40   </script>
41 </head><body></body>
42 </html>
```

上午 12:02:36
星期二 2021年3月30日

Using Relational Opera

file:///D:/D槽/Dropbox/105下(網路程式設計)/Slides_第

Good Morning, Jerry, welcome to JavaScript programming!

(Ex: 延伸應用：在不同時間，播放不同背景音樂)

6.7 Decision Making: Equality and Relational Operators (Cont.)



- ▶ **Strict Equals** (“===”) and **Strict Does Not Equal** (“!==”) Operators 嚴格等式判斷
 - The comparison “75” === 75 yields the value **false** (不成立) because one operand is a **string** (字串) and the other is a **number** (數字).
 - 註：型態不相等就會不成立
(字串型態) (數字型態)
 - Similarly, “75” !== 75 yields **true** (成立) because the operand’s **types are not equal** (因為型態不相等), therefore the values **are not equal**.

Operators	Associativity	Type
* / %	left to right	multiplicative
+ -	left to right	additive
< <= > >=	left to right	relational
== != === !==	left to right	equality
=	right to left	assignment

Fig. 6.15 | Precedence and associativity of the operators discussed so far.

In-Class Exercise #1

- ▶ 課中練習：請同學利用JavaScript建立一個網頁，當開啟網頁時：
 - 1) 透過文字對話框，詢問使用者名稱
 - 2) 根據目前時間依早上(00:00~12:00)、中午(12:00~18:00)、晚上(18:00~24:00)
 - 1) 更換不同圖片或背景 (補充：<body background="背景圖片網址"> 方式不限於此)
 - 2) 使用不同顏色文字問候
Ex: Good morning、Good afternoon、Good evening
- Note (範例):
 - 字體顏色： ... ，其他顏色如：'red', 'blue', 'green'
 - 設定背景顏色：<body bgcolor="#cccccc">, <body bgcolor="orange">
 - 設定背景圖片：<body background="背景圖片的網址">
 - (作法無限制，亦可查網路資訊)

(可參考FTP範例程式，有類似程式)



(延伸)

Chapter 7

JavaScript: Control Statements, Part 1

Internet & World Wide Web
How to Program, 5/e

<ftp://111webdesign:webdesign@120.114.190.22:4000>



OBJECTIVES

In this chapter you will:

- Learn basic problem-solving techniques.
- Develop algorithms through the process of top-down, stepwise refinement.
- Use the `if` and `if...else` selection statements to choose among alternative actions.
- Use the `while` repetition statement to execute statements in a script repeatedly.
- Implement counter-controlled repetition and sentinel-controlled repetition.
- Use the increment, decrement and assignment operators.

(下週)

7.2 Algorithms (演算法)

- ▶ Any **computable problem** (可計算的問題) can be solved by **executing a series of actions** in a specific order (一序列的動作)
 - 註：任何可由計算完成的問題，都可以透過執行一序列的動作來解決
- ▶ **Algorithm** (演算法)
 - A **procedure** (程序) for solving a problem in terms of
 - the **actions** (動作) to be executed,
 - the **order** (順序) that the **actions** are to be executed
- ▶ **Pseudocode** (虛擬程式碼)
 - An **informal language** (非正式程式語言) that **helps** users **develop** algorithms
 - **Similar to English** (白話文)
 - Very **convenient** (方便構思、打草稿) and **user friendly** (友善、直覺)

7.4 Control Statements

- ▶ **Sequential execution** // 順序性的執行
 - Execute statements **in the order** they appear in the code
 - 註：根據程式碼出現的順序，執行statement
- ▶ All scripts can be **written in three** control statements (**3類控制陳述**)
 1. Sequence (順序)
 2. Selection (條件，如：if.. else)
 3. Repetition (迴圈，如：while..., for...)

(flowchart流程圖：用於方法表達，很重要)

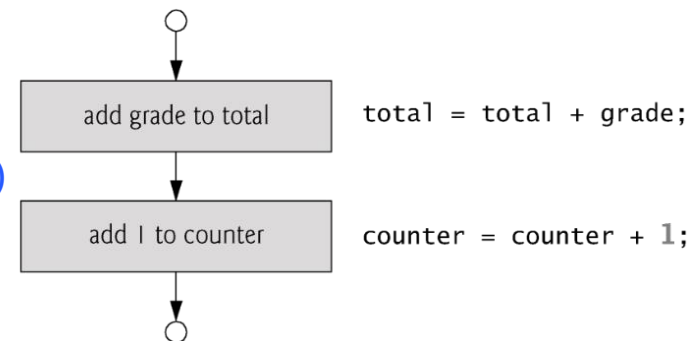


Fig. 7.1 | Flowcharting JavaScript's sequence structure.

7.4 Control Statements (Cont.)

▶ Flowchart 流程圖

- A **graphical** (圖形化) representation of an algorithm
- Using certain **special symbols** (特殊圖案/符號),
 - such as **rectangles** (矩形), **diamonds** (菱形), **ovals** (橢圓形) and **small circles** (小圓圈).

▶ Oval symbols (橢圓形)

- ▶ Represent the “**Begin** (開始)” and “**End** (結束)” of the **algorithm**.
- ▶ For a **portion of an algorithm** (片段演算法), the oval symbols are **replaced by** (取代) **small circle symbols** (小圓圈).
- ▶ 註：若只描述**部分algorithm**，則使用**小圓圈**表示**起始及結束**。

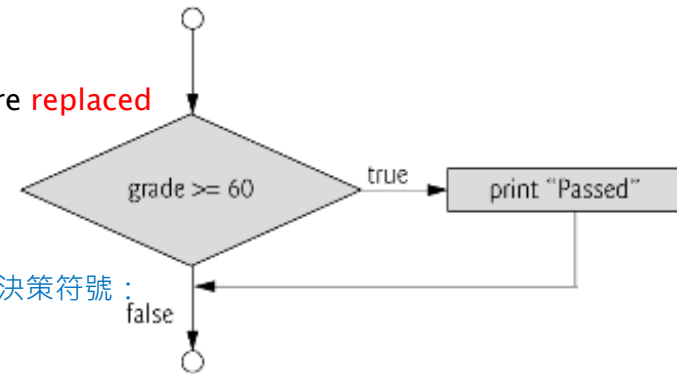
▶ Diamond symbol (菱形)

- ▶ The **most important** symbol, also called the **decision symbol** (決策符號：條件判斷), which indicates to **make a decision** (作判斷).

▶ Rectangle symbols (矩形)

- ▶ Indicates actions, including a **calculation** (執行計算) or an **input/output operation** (讀取/寫入).

- Symbols are connected by **arrows** (箭頭) called **flowlines** (流程線),
 - which indicate **the order** (順序) of **executing actions**





7.4 Control Statements (Cont.)

- ▶ JavaScript provides **three selection structures** (選擇結構).
 - The **if ...** statement (單選陳述，沒有else)
 - **performs an action** if a **condition is true** or **skips** the action if the condition is false.
 - Called a **single-selection statement** (單選陳述：做或不做) because it **selects** or **ignores** a single action or group of actions.
 - 註：if ... 為單選陳述，條件符合即**做**，不符合則**不做**
 - The **if...else...** statement (雙選陳述)
 - **performs an action** if a **condition is true** and **performs** a **different** action if the **condition is false**.
 - **Double-selection statement** (雙選陳述) because it selects between **two different actions** or group of actions.
 - 註：if...else... 為雙選陳述，條件符合即**做A**，不符合則**做B**
 - The **switch ...** statement (多選陳述)
 - **performs one of** many different actions, depending on the **value** of an expression.
 - **Multiple-selection statement** (多選陳述) because it selects among many different actions or groups of actions.
 - 註：**switch...** 為多選陳述，依符合哪一條件則**做哪一動作**

(switch用法)

```
switch (...)  
{  
    case '...':  
        ... //動作  
        break;  
    case '...':  
        ... //動作  
        break;  
    default:  
        ... //動作  
        break;  
}
```

7.4 Control Statements (Cont.)

- ▶ JavaScript provides **four** repetition statements:
 - while, do...while, for, for...in.

(詳細下週介紹)

JavaScript
保留字詞

JavaScript reserved keywords

break	case	catch	continue	default
delete	do	else	false	finally
for	function	if	in	instanceof
new	null	return	switch	this
throw	true	try	typeof	var
void	while	with		

Keywords that are reserved but not used by JavaScript

class	const	enum	export	extends
implements	import	interface	let	package
private	protected	public	static	super
yield				

Fig. 7.2 | JavaScript reserved keywords.

7.5 if ... Selection Statement

- ▶ The JavaScript interpreter ignores *white-space characters* (空白字元)
 - Blanks (空白鍵), tabs (Tab鍵) and newlines (換行) used for indentation (縮排)

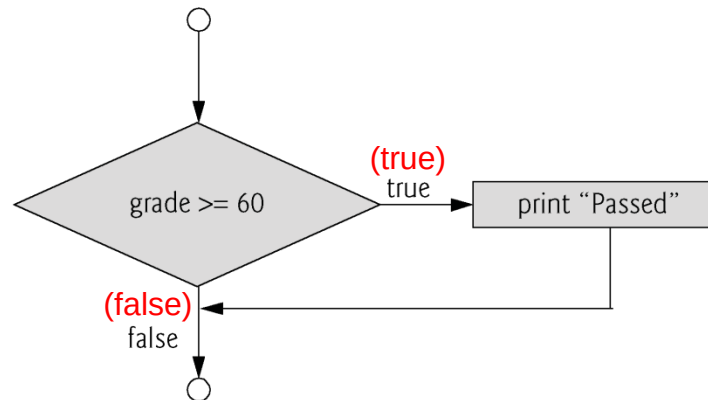


Fig. 7.3 | Flowcharting the single-selection if statement.

7.6 if...else Selection Statement



- ▶ To specify that **two different actions** should be performed when the condition is **true** and when the condition is **false**.

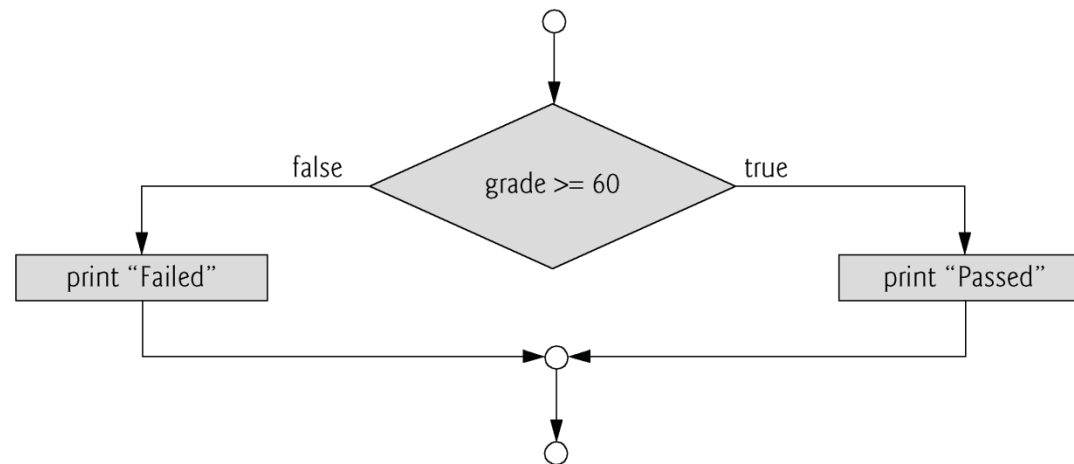


Fig. 7.4 | Flowcharting the double-selection if...else statement.

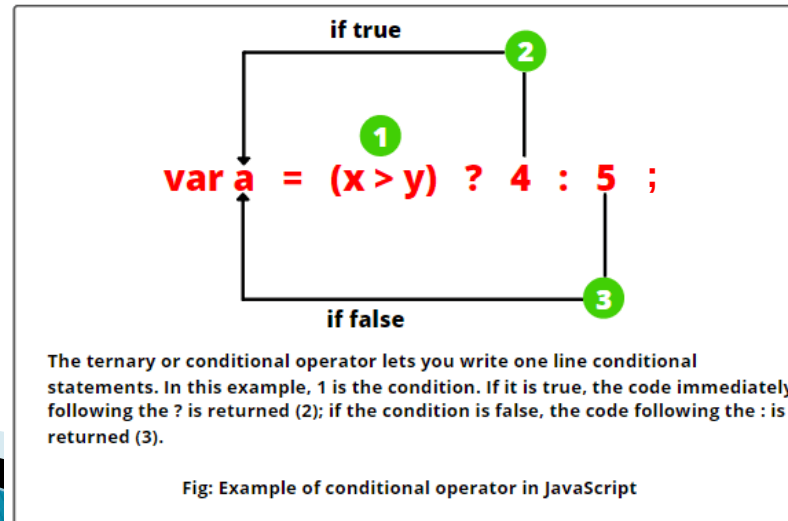
7.6 **if... else...** Selection Statement (Cont.)



條件 成立=>做 不成立=>做

▶ **Conditional operator** 更精簡 (... ? ... : ...)

- 註：等同於 **if...else...** statement
- it includes **three operands** (含三個運算元)
 - The **first operand** is a **boolean expression** (條件式)
 - The **second** is **the action** (成立，欲執行的動作) if the boolean expression evaluates to **true**
 - The **Third** is **the action** (不成立，欲執行的動作) if the boolean expression evaluates to **false**



7.6 **if...else** Selection Statement (Cont.)

- ▶ Nested **if...else...** statements **巢狀條件式**
 - for **multiple cases** by placing **if...else...** statements **inside** other **if...else...** statements
 - 註：**else**沒註明，則會關聯至最近的**if**
 - 註：建議用**大括弧 {...}** 避免**關聯錯誤**
 - **Block (區塊)**: a **set of statements** contained within a **pair of braces {...}**

```
if ( x > 5 )  
    if ( y > 5 )  
        document.writeln( "<p>x and y are > 5</p>" );  
else  //else沒註明，則會關聯至最近的if  
    document.writeln( "<p>x is <= 5</p>" );
```



```
if ( x > 5 )  
{  
    if ( y > 5 )  
        document.writeln( "<p>x and y are > 5</p>" );  
}  
else  
    document.writeln( "<p>x is <= 5</p>" );
```

- **fatal logic error**: causes a script to fail and terminate prematurely.
 - 註：會導致程式**crash**的錯誤。
- **nonfatal logic error**: allows a script to **continue executing**, but the **script produces incorrect** results.
 - 註：**可執行**，但**結果**不正確 (類似**logic error**)。



Lab

(上機練習)

In-Class Exercise #1

- ▶ 課中練習：請同學利用JavaScript建立一個網頁，當開啟網頁時：
 - 1) 透過文字對話框，詢問使用者名稱
 - 2) 根據目前時間依早上(00:00~12:00)、中午(12:00~18:00)、晚上(18:00~24:00)
 - 1) 更換不同圖片或背景 (補充：<body background="背景圖片網址"> 方式不限於此)
 - 2) 使用不同顏色文字問候
Ex: Good morning、Good afternoon、Good evening
- Note (範例):
 - 字體顏色： ... ，其他顏色如：'red', 'blue', 'green'
 - 設定背景顏色：<body bgcolor="#cccccc">, <body bgcolor="orange">
 - 設定背景圖片：<body background="背景圖片的網址">
 - (作法無限制，亦可查網路資訊)

(可參考FTP範例程式，有類似程式)

In-Class Exercise #2

- ▶ **課中練習**：請同學利用對話框的方式，簡單設計一個猜數字的網頁小遊戲，只給使用者**2次(或3次)**機會：
 - 數值**正確**：顯示“**Bingo!**”
 - 數值**過小**：顯示“**請增大數字!**”
 - 數值**過大**：顯示“**請減小數字!**”
 - Ps.
 - 因為還未教到**迴圈**，可利用多組 **if...else...** 方式，提供使用者**2次(或3次)**機會
 - **隨機數**尚未教到，如欲使用，可練習查詢Google，關鍵字，例如：**javascript 隨機數**。
 - 善用chrome瀏覽器的**F12**，可快速檢查語法是否有錯誤（如有錯誤，會出現  ）

(★提醒: 對話框取得輸入文字後，請記得轉成數字型態parseInt，才能進行運算)

- **注意：**

23:59 前上傳到ftp (否則視為遲交)：

<ftp://111webdesign:webdesign@120.114.190.22:4000>

- **Lab05**

- **請以“學號”建立資料匣 (否則扣分)**



【工商服務】

老師FB分享：學長姐發展消息



Jia-Ming Liang 😊 覺得開心——和 Kris Hung 及其他 2 人，在桃園市桃園區。

2020年5月8日 · 桃園市桃園區 · 🌐

恭喜~😊🎉 語歆 (UC Irvine)、和寰 (Rice university)、德融 (NYU Tandon)~錄取美國研究所~!! 🎓🎓🎓

擔任你們的推薦人，老師也感到非常光榮哦~!! 希望未來能有機會回國貢獻所學~! 😊👍👍... 查看更多



May. 08. 2020



Jia-Ming Liang ——和蔡宛凌及其他 4 人，在聯華神通集團總部。

4月16日 · 台北市 · 🌐

恭喜 ~!! 宛凌、志豪、胤恩、香伶~ 🎉

我們專題「智慧水族箱」獲得 2021育秀盃 優選~! 🏆

#開心做專題

#學生棒棒... 查看更多



老師FB分享：學長姐專題成果



Jia-Ming Liang 和賴怡誠及其他 4 人。

2020年11月6日 · 2人

恭喜~ 怡誠, 冠綸, 峻瑋, 淵翔~! 🎉🎉🎉

我們專題『智慧行動商店服務』，獲得2020全國大專校院智慧創新競賽 ~ 第2名耶🏆~!! 🎉🎉

#謝謝評審
#獎金5萬元
#繼續努力
#遲來的祝福



Jia-Ming Liang 😊 覺得感恩—— 和 James Chen 及其他 5 人。

2022年6月3日 · 2人

恭喜~ 羽柔、郁雯、孟軒、德帆、昀昀~!



大家一起完成的「智慧點滴系統」，共獲得了大專智創、三創競賽等前三名~!🏆 及最佳論文獎~!🏆 ... 查看更多

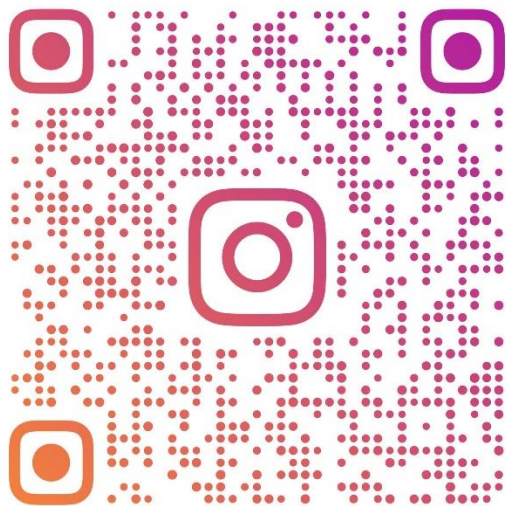




【工商服務】❤️按讚追蹤南大電機IG粉絲頁

最新消息

- 國立臺南大學電機工程學系
- IG粉絲頁
- https://www.instagram.com/nutn_ee/?hl=zh-tw



NUTN_EE

國立臺南大學電機工程學系
4月26日 · 🌐

賀！本系學士班110級考取碩士班榜單！

國立臺南大學電機工程學系 110級考取碩士班榜單	
栢德驊	推甄 國立陽明交通大學光電工程學系(備取已錄取) 國立成功大學工程科學系(備取已錄取) 國立成功大學電機工程學系(備取已錄取) 國立成功大學奈米精體電路工程碩士學位學程(備取已錄取)
翁晉苙	推甄 國立臺灣大學光電工程學研究所(備取已錄取) 國立陽明交通大學光電工程學系(備取已錄取) 國立中山大學電機工程學系(正取) 國立清華大學光電工程研究所(備取)
張恩誠	推甄 國立臺灣大學光電工程學研究所(備取已錄取) 國立臺灣大學工程科學及海洋工程學系(備取已錄取) 國立成功大學工程科學系(正取) 國立成功大學光電科學與工程學系(備取已錄取) 國立成功大學電機工程學系(備取已錄取)
陳盈豪	推甄 國立陽明交通大學智慧系統與應用研究所(備取已錄取) 國立陽明交通大學智慧計算與科技研究所(備取已錄取)

國立臺南大學電機工程學系
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國立成功大學工程科學系(備取已錄取)
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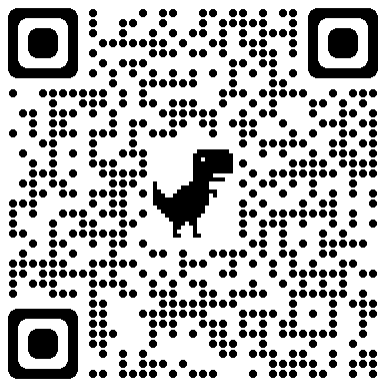
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國立成功大學光電科學與工程學系(備取已錄取)
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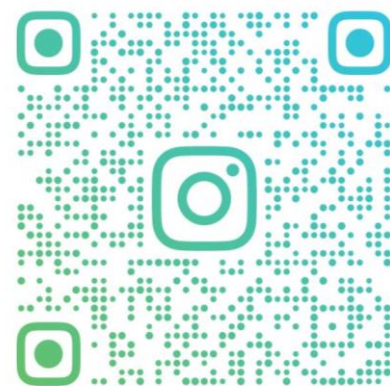


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世界很大～帶著夢想去看看～!

Nutn.EE 家銘老師